BFW STEAM HEATER VER 1

Operator Training Simulator · E-101

Chulalongkorn University · Department of Chemical Engineering

EQUIPMENT LIST AND DATASHEETS

DOCUMENT NO. : BFWH-V1-ME-EQL-001

DISCIPLINE : Mechanical

PROJECT : BFW Steam Heater Ver 1 · E-101 Operator Training Simulator

STATUS : Issued for teaching (IFT)

Equipment list of the BFW steam heater unit with design pressures and temperatures, and datasheets for E-101 (TEMA type, thermal rating, differential expansion, GA sketch, nozzles, UG-27 wall thickness, test pressures, vacuum), P-101A/B, TK-101, Y-101/ST-101 and VB-101. Design conditions checked against the model relief runs. Assumptions are labelled.

Rev	Date	Description	Originator	Checker	Approver
0	25-Oct-26	Issued for teaching	A. Soottitantawat	—	—

Numbers are taken from the BFW Steam Heater Ver 1 simulator model; values marked assumed / TBC / HOLD are not design data.

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1 Purpose and basis

This document lists every tagged item and gives the datasheet the vendor quotes against. Process inputs are from BFWH-V1-PE-HMB-001; E-101 sizing from the Design tab; relief cases from the model relief runs.

Design pressures: shell 350 kPa g = PSV-100 set; tube side 1,100 kPa g ≥ pump shut-off 1,030 kPa g (both **assumed**, Design tab). Design temperature 180 °C = relieving temperature 151.1 °C (saturation at 385 kPa g) + 25 K, rounded up to 5 °C (**assumed**). FV = full vacuum. Weights, tubesheet and expansion joint are HOLD.

2 Equipment list

Tag	Description	Type	Size / capacity	Design shell P / T	Design tube / casing P / T	Material	Driver	Weight
E-101	BFW steam heater	Shell and tube, horizontal, TEMA BEM, fixed tubesheet (assumed)	6,040 kW; 46.8 m²; 100 tubes × 9.39 m; shell ID 390 mm	350 kPa g / FV TBC; 180 °C	1,100 kPa g; 180 °C	Shell, bonnets SA-516-70; tubes SA-179 (assumed)	—	HOLD
P-101A/B	BFW pumps (1 running, 1 spare)	Centrifugal, horizontal end-suction (assumed)	123.6 m³/h at 55.3 m; shut-off 102.1 m	—	casing 1,100 kPa g; 60 °C (assumed)	CS casing, stainless impeller (assumed)	Motor ≥ 30.6 kW	HOLD
TK-101	BFW storage tank	Vertical cylindrical, atmospheric, vented (assumed)	HOLD (indicative 20.6 m³, 10 min of rated flow, assumed)	Atmospheric + liquid head; 60 °C (assumed)	—	CS, internally coated (assumed)	—	HOLD
Y-101	Condensate strainer	Y-type, 3", 0.8 mm screen (assumed)	3"	350 kPa g; 180 °C	—	CS body, SS screen (assumed)	—	HOLD
ST-101	Steam trap	Float and thermostatic (assumed)	24,838 kg/h at ΔP 75 kPa (SF 2.5)	350 kPa g; 180 °C	—	CS body, SS internals (assumed)	—	HOLD
VB-101	Vacuum breaker	Spring-loaded, opens at −2.5 kPa g (model)	1" (assumed); capacity HOLD	350 kPa g; 180 °C	—	Stainless steel (assumed)	—	HOLD
PSV-100	Shell relief valve	Spring-loaded relief valve	Size HOLD (Relief & vent)	set 350 kPa g (assumed, TBC)	—	CS body (assumed)	—	HOLD

All items are on BFWH-V1-PID-001 rev D.

3 E-101 datasheet

TEMA BEM (assumed). B bonnet front and M bonnet rear: one tube pass needs BFW in at one end and out at the other; bonnets are lifted off to clean the tubes from inside, the side that scales. E one-pass shell: steam enters near the front and non-condensables are swept to the far end, where the vent is. Fixed tubesheets suit clean condensing steam on the shell; the cost is differential expansion (below). TEMA class R (assumed).

Thermal rating (Design tab)

Item	Value	Note
Duty Q	6,040.1 kW	$\dot{m}c_p\Delta T$, 100.0 m³/h, 18.0 → 70.0 °C
Steam (shell)	9,935 kg/h at 105.0 kPa g / 121.3 °C	λ 2,188.6 kJ/kg
LMTD	74.31 K	isothermal shell, F = 1
U service / clean	1,736 / 3,764 W/m² K	h_i 10,787, h_o 8,958 W/m² K
Fouling BFW / steam side	0.000176 / 8.8e-05 m² K/W	assumed (Design tab)
Area A = Q/(U·LMTD)	46.81 m² outside	clean overdesign 117 %
Tubes	100 × 15.875 × 1.651 mm (16 BWG), 1 pass	ID 12.57 mm; velocity 2.24 m/s; Re 45,242
Tube length	9.39 m	≤ L_{max} 12.0 m (assumed)
Pitch	19.84 mm triangular	1.25 d_o
Bundle / shell ID	232 / 386 mm → 390 mm	rounded up (assumed)
Tube-side ΔP	60.5 kPa	model PDI-109 60 kPa
Steam inlet nozzle	321 mm required at 30 m/s (assumed)	→ 14" sch 40, ID 333.3 mm
Shell volume	1.10 m³	π/4 D_s^2 L

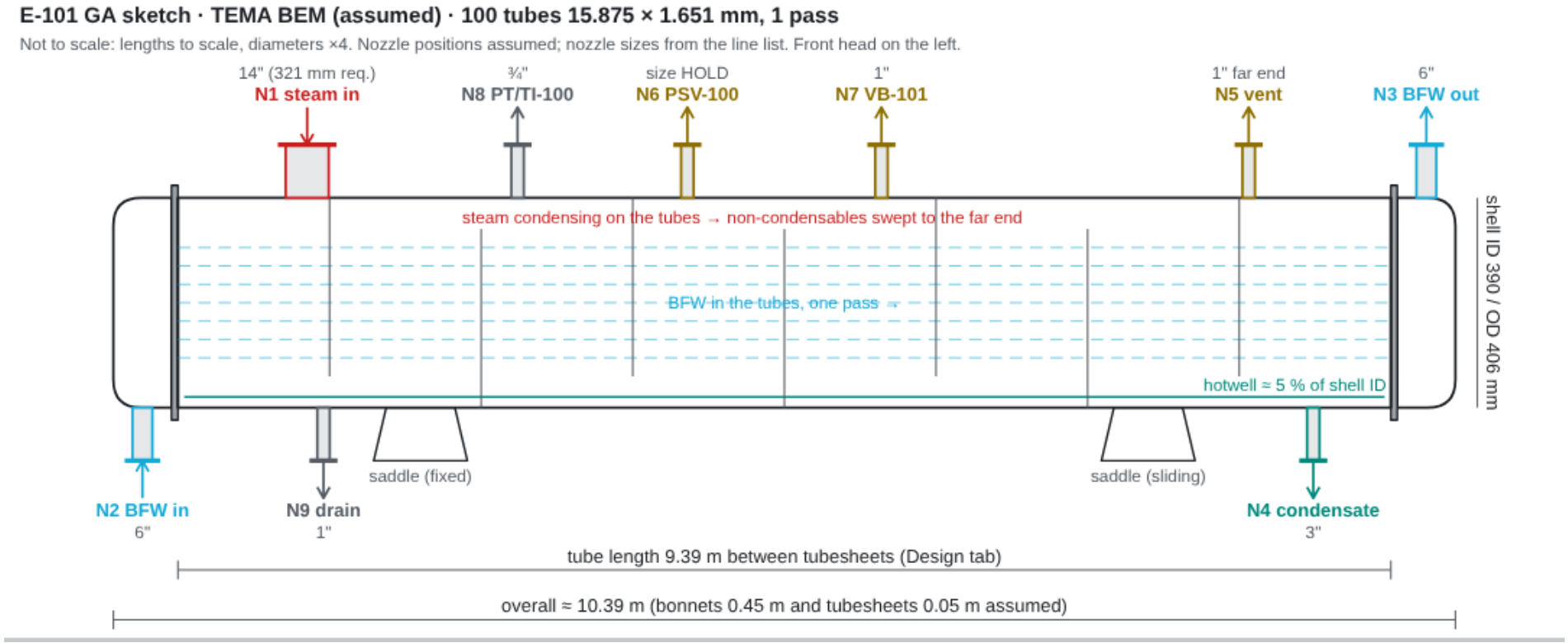
Nozzle schedule

Nozzle	Service	NPS	ID mm	Location (assumed)
N1	Steam inlet (14"-LS-102)	14"	333.3	shell top, 1.0 m from the front tubesheet; impingement plate
N2	BFW inlet (6"-BW-101)	6"	154.1	front bonnet, bottom
N3	BFW outlet (6"-BW-103)	6"	154.1	rear bonnet, top
N4	Condensate outlet (3"-SC-101)	3"	77.9	shell bottom, 0.6 m from the rear tubesheet (hotwell)
N5	Air vent (1"-VT-101, V-104)	1"	26.6	shell top, far end from the steam inlet
N6	PSV-100 inlet	HOLD	—	shell top, mid-length; size from the Relief & vent tab
N7	VB-101 vacuum breaker	1"	26.6	shell top
N8	PT-100 / TI-100	0.75"	20.9	shell top
N9	Shell drain (V-105)	1"	26.6	shell bottom, front end

Differential expansion: T_{tube} = 44.0 + 0.621 × (121.3 − 44.0) = 92.0 °C; Δ = αLΔT = 1.2e-05 × 9,386 × 29.3 = **3.3 mm** at design, **11.6 mm** at start-up (tubes at 18 °C). Expansion joint HOLD — EQ-22.

Flanges ASME B16.5 Class 150 (assumed): group 1.1 carbon steel about 1,460 kPa g at 180 °C (typical).

4 GA sketch



E-101 GA sketch, not to scale (lengths to scale, diameters ×4); nozzle positions, bonnet and tubesheet lengths assumed.

5 Wall thickness, test pressures and vacuum

ASME VIII-1 UG-27 circumferential stress: t = PR/(SE − 0.6P) + CA, R inside radius in the corroded condition. MAWP from the available wall (nominal less mill tolerance less CA).

Shell worked: t = 0.350 × 198.00 / (138 × 0.85 − 0.6 × 0.350) + 3 = 0.592 + 3 = **3.59 mm** ≤ 8 mm.

- Hydrotest: shell 1.3 × 350 = **455 kPa g**; tube side 1.3 × 1,100 = **1,430 kPa g** (stress ratio 1, assumed).
- Full vacuum, indicative long cylinder: $p_c = 2E/(1 - \nu^2)(t/D_o)^3$ = 821 kPa with t 5.0 mm corroded, D_o 406 mm; ÷3 = 274 kPa ≥ 101.3. UG-28 and FV stamp HOLD — EQ-23. VB-101 opens at −2.5 kPa g (HAZOP R14).

Component	P kPa g	D_i mm	S MPa	E	CA mm	R mm	t_P mm	t + CA mm	t nom mm	tol %	MAWP kPa g	OK
Shell (SA-516-70)	350	390.00	138.0	0.85	3.0	198.00	0.592	3.59	8.000	0	2,918	✓
Bonnets / channel (SA-516-70)	1,100	390.00	138.0	0.85	3.0	198.00	1.867	4.87	8.000	0	2,918	✓
Tubes (SA-179)	1,100	12.57	92.4	1.00	0.0	6.29	0.075	0.08	1.651	10	19,127	✓

Assumed: SA-516-70 S 138 MPa at 180 °C, E 0.85 (spot RT), CA 3 mm, plate 8 mm; SA-179 tubes S 92.4 MPa, E 1, CA 0, 16 BWG with 10 % under-tolerance. Stamped MAWP taken = design pressure (assumed).

6 P-101A/B datasheet

Liquid BFW 18 °C (cold case 10.7 °C), 998 kg/m³, vapour pressure 2.06 kPa abs. Centrifugal, horizontal end-suction, 2-pole, mechanical seal (assumed).

Item	Normal	Rated	Note
Flow incl. recycle, m³/h	110.7	123.6	rated = maximum-load case (BFWH-V1-PE-HMB-001)
Suction / discharge, kPa g	30 / 662.4	30 / 571.6	model curve; PI-106
Differential kPa / head m	632.4 / 64.6	541.6 / 55.3	H = ΔP/ρg
Shut-off	1,030 kPa g / 102.1 m	—	H_0 1000 kPa (model)
NPSHa, m	13.01	12.96	NPSHr 4 m assumed, HOLD (vendor)
Hydraulic / absorbed power, kW	19.4 / 27.8	18.6 / 26.6	η 0.7 (assumed)
Motor	≥ 30.6 kW	—	1.15 × rated absorbed (assumed); rating on the electrical load list
Minimum flow	FV-101B recycle, up to 29.5 m³/h	—	pump minimum continuous flow HOLD; fixed recycle line (HAZOP R03)
Casing design	—	1,100 kPa g, 60 °C (assumed)	≥ shut-off 1,030 kPa g

7 TK-101, Y-101/ST-101 and VB-101 datasheets

TK-101 BFW storage tank

Item	Value
Type	Vertical cylindrical, flat bottom, fixed roof, vented (assumed)
Operating / design	atmospheric, 18 °C / atmospheric + liquid head, 60 °C (assumed)
Working volume	HOLD — indicative 20.6 m³ = 10 min × 123.6 m³/h (assumed)
Diameter × height	HOLD
Minimum level above pump centreline	3.06 m (= K.Ptk 30 kPa, assumed)
Nozzles	outlet 8"; recycle return 3"; make-up, overflow, vent HOLD
Instruments	LT-101 recommended (HAZOP R02)

Y-101 strainer and ST-101 steam trap

Item	Value	Note
Required capacity	24,838 kg/h at ΔP 75 kPa	SF 2.5 × design 9,935 kg/h (start-up)
Model trap capacity at design ΔP	14,840 kg/h	$K_{trap}\sqrt{\Delta P}$ = 0.476 × √75 × 3600 = 1.49 × design → EQ-21
At maximum load	19,425 kg/h at ΔP 128.5 kPa	load 12,161 kg/h ✓
At low load (stall)	6,876 kg/h at ΔP 16.1 kPa	load 7,074 kg/h — trap capacity sets the stall point
Max inlet / back-pressure	350 / 30 kPa g	shell design; return header

Item	Value	Note
Type	Float and thermostatic (assumed)	continuous discharge; thermostatic air vent
Strainer Y-101	3", 0.8 mm perforated screen, blow-down (assumed)	clean after maintenance (HAZOP R15)

VB-101 vacuum breaker

Item	Value
Set	opens at −2.5 kPa g (model)
Size	1" (assumed); capacity HOLD (HAZOP R14)
Location	shell top (N7)
Material	stainless steel (assumed)

8 Checks

Model relief runs (trips disabled)

Run	Peak shell kPa g	Max BFW out °C	In fault range
FV-104 fails open, header at design	166.8	74.3	yes
FV-104 fails open, header high (300 kPa g)	267.0	79.9	yes
FV-104 fails open, header let-down failure (toward 400 kPa g, beyond the fault range)	367.0	84.3	no — PSV-100 case
Condensate outlet blocked (strainer/trap fully plugged)	198.9	70.0	yes
Condensate return back-pressure at maximum (160 kPa g)	181.8	70.0	yes
Tube leak into the shell (1.5 kg/s)	118.7	70.0	yes
BFW flow lost with steam held (pump trips, spare unavailable)	200.0	126.0	yes

Checks

Check	Values	Result
Shell design ≥ highest credible shell pressure (relief runs)	350 ≥ 267.0 kPa g (FV-104 fails open, header high (300 kPa g))	✓
Shell design ≥ steam header maximum	350 ≥ 300 kPa g (assumed)	✓
Beyond the fault range: PSV-100 set ≤ shell design	peak 367.0 kPa g; set 350 kPa g	✓
Shell design T ≥ relieving temperature	180 ≥ 151.1 °C	✓
Tube design T ≥ hottest BFW in the relief runs	180 ≥ 126.0 °C	✓
Tube design ≥ pump shut-off	1,100 ≥ 1,030 kPa g	✓
Tube length ≤ maximum	9.39 ≤ 12.0 m	✓
Rating = Design tab sizing; UA and duty = model	A 46.81 m², UA 81.28 kW/K, Q 6,040.1 kW	✓
Tube count = P&ID title block	100 tubes, 1 pass	✓
Shell ID ≥ Design tab; N1 ≥ required; BFW nozzles = Design tab	390 ≥ 385.8; 333.3 ≥ 321.5; 154.1 mm	✓
Shell, bonnet and tube walls (UG-27)	3.59, 4.87, 0.084 mm required	✓
Flange Class 150 ≥ tube-side design (assumed)	1,460 ≥ 1,100 kPa g	✓
Full vacuum (indicative)	274 ≥ 101.3 kPa	✓
Pump normal point = PI-106 and FIC-101 + recycle	662.4 kPa g, 110.7 m³/h	✓
Model trap capacity ≥ datasheet requirement (SF 2.5)	14,840 vs 24,838 kg/h	✗ EQ-21 raised
Shell design pressure agrees with the LOPA tab	350 vs 300 kPa g	✗ EQ-20 raised
P-101 motor basis agrees with the electrical load list	≥ 30.6 vs 32.1 kW (37 kW motor)	✗ EQ-24 raised
Every listed tag appears on BFWH-V1-PID-001 rev D	E-101, P-101A, P-101B, TK-101, Y-101, ST-101, VB-101, PSV-100	✓

9 Engineering queries

EQ	Subject	Query	Status
EQ-20	E-101 shell design pressure 350 vs 300 kPa g	The Design tab sets the shell design pressure at 350 kPa g (= PSV-100 set, assumed) and this equipment list uses it; the LOPA tab (bow-tie BT1) and the Safety layers tab use a shell rated 300 kPa g (assumed). Reconcile to one value. At 300 kPa g the shell design equals the assumed header maximum, leaving no margin for the FV-104-fails-open case at the header maximum (model 267 kPa g) plus PSV accumulation.	Open

EQ	Subject	Query	Status
EQ-21	Steam trap ST-101 capacity	The datasheet requires 24 838 kg/h at 75 kPa (SF 2.5 on the design condensate, start-up with a cold shell). The trap in the simulator model passes 14 840 kg/h at 75 kPa (1.49 × design). Confirm the trap selection with the vendor, or accept a lower safety factor with the slow warm-up in the start-up procedure; the model then represents a smaller trap than the datasheet.	Open
EQ-22	Fixed-tubesheet differential expansion	Shell at 121.3 °C and tube metal at about 92 °C give 3.3 mm differential expansion over 9.39 m at design and 11.6 mm at start-up (tubes at 18 °C, shell hot). A fixed-tubesheet BEM needs a shell expansion joint or a tubesheet design (UHX) that takes the load; alternatives are a floating head or U-tubes (two passes, changes the thermal design). Expansion joint and tubesheet HOLD.	Open
EQ-23	E-101 shell full-vacuum rating	After a trip the shell condenses toward the saturation pressure of the BFW (near full vacuum). An indicative long-cylinder check gives 274 kPa allowable for an 8 mm shell, above 101.3 kPa, but the UG-28 calculation and the FV stamp are not done. Confirm the FV rating so protection does not rely on VB-101 alone (HAZOP R14).	Open
EQ-24	P-101 motor rating basis differs between Mechanical and Electrical	This equipment list sizes the P-101A/B motor on 1.15 × the absorbed power at the rated point (26.6 kW at 123.6 m³/h) = at least 30.6 kW. The electrical load list (BFWH-V1-EE-LL-001) sizes it on 1.15 × the peak of the absorbed-power curve (27.89 kW near 105 m³/h; maximum-load case 125.6 m³/h) = 32.1 kW and selects a 37 kW motor. Agree one sizing basis (rated point or end of curve / maximum flow, and the margin) and update both documents; the pump datasheet should state the motor rating.	Open

Notes

- Numbers are generated from the simulator model (data.json of the Mechanical tab); the tab and this document share them.
- Weights, tubesheet thickness, expansion joint, PSV-100 size, NPSHr and TK-101 dimensions are HOLD.
- Do not use for construction.